

Multi-task Learning-based Text Classification with Subword-Phrase Extraction

Yusuke Kimura
kimura@mil.doshisha.ac.jp
Doshisha University
Japan

Takahiro Komamizu
taka-coma@acm.org
Nagoya University
Japan

Kenji Hatano
hatano@acm.org
Doshisha University
Japan

ABSTRACT

Text classification using deep learning, which is trained with a tremendous amount of text, has achieved superior performance than traditional methods. In addition to its success, multi-task learning has become a promising approach for text classification; for instance, a multi-task learning approach employs named entity recognition as an auxiliary task for text classification. The existing MTL-based text classification methods depend on auxiliary tasks using supervised labels, which require large human and/or financial efforts to create. To reduce these efforts, this paper proposes a multi-task learning-based text classification framework which reduces the additional efforts on supervised label creation. A basic idea to realize this is that to utilize phrasal expressions consisting of subwords (called subword-phrase). To the best of our knowledge, there has been no text classification approach on top of subword-phrases, because subwords do not always express a coherent set of meanings. The proposed framework is new to add subword-phrase recognition as an auxiliary task, and to utilize subword-phrases for text classification. To realize the low-cost auxiliary recognition task, the framework extracts subword-phrases in an unsupervised manner. The experimental evaluation of the five popular datasets for text classification showcases the effectiveness of the involvement of the subword-phrase recognition as an auxiliary task. It also shows comparative results with the state-of-the-art method.

CCS CONCEPTS

• **Computing methodologies** → **Classification and regression trees**; *Natural language processing*; Neural networks.

KEYWORDS

Text Classification, Subword-phrase, Multi-task Learning

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1 INTRODUCTION

Text classification is a fundamental technology that can be used for various applications, providing effective and efficient information access to a massive amount of text [10, 21]. Text classification has been used for large variety of applications such as detecting hate speech [7], categorizing daily news articles, unfair clause detection in terms of services [13], and so forth. Text classification is a supervised classification task, meaning that a human-labeled dataset that reflects a criterion for classifying texts is required for each specific task. A classifier learns classification criteria in a feature space based on the dataset. Traditionally, hand-crafted features (e.g., term frequency-inversed document frequency) have been used for text classification. In recent literature, deep learning-based technologies have achieved higher classification performance than ever. One of the successful components of these technologies is pre-trained models (like BERT [8]) trained with a vast amount of text. It provides semantically rich features for texts; therefore, even a simple multi-layer perceptron-based classifier can achieve better performance. After the first success, many pre-trained models have been published, such as RoBERTa [17] and GPT-3 [5].

In these pre-trained models, subword-based tokenization [26] has become popular, meaning that texts are divided into subwords which are smaller units than words. Using subwords minimizes the presence of unknown tokens and the ability to represent texts by limited vocabulary size. Although the subword-based tokenization has started for machine translation [26], it is used in various natural language processing tasks including text classification.

Multi-task learning (MTL for short) [6] is a promising approach to lift up performances of deep learning models, which involves one or more auxiliary tasks with the main task by sharing parameters. It has also been applied to text classification [15, 30, 31]. Learning models with auxiliary tasks positively affects the generalization performance of the main task and reduces over-fitting. Early work on MTL-based text classification [15, 30] focuses on the way to combine multiple tasks and combined tasks in different datasets. Recent work combines text classification with auxiliary tasks in the same dataset, such as named entity recognition (NER for short) [2, 28] or label co-occurrence prediction [31].

The success of NER-based MTL approaches suggests that recognizing phrasal expressions like named entities can be a suitable auxiliary task for MTL-based text classification. However, to realize NER as an auxiliary task for MTL-based text classification, supervised labels for NER are required in addition to those for text classification, and constructing such datasets is costly because additional expensive human efforts are needed for labeling for NER.

Therefore, in this paper, we seek the possibility of MTL-based text classification with phrasal expression recognition which does

not require additional human effort to construct a training dataset. Phrasal expressions (or key phrases) for texts have been studied in past decades [24, 32]. Key phrase extraction is not straightforward to apply based on the tokenization of popular pre-trained neural language models. Because the granularity of tokens is mismatched, meaning that the tokenization is based on subwords rather than words. Therefore, this paper defines a phrasal expression based on subwords as a *subword-phrase*. In contrast with phrasal expressions based on words, subword-phrases are not necessarily semantically coherent because a vocabulary of subwords is determined statistically [26]. Due to this less semantically coherence of subword-phrases, there has been no work using subword-phrases.

In this paper, we propose a framework of MTL-based text classification with subword-phrase recognition to improve the accuracy of text classification. Our framework consists of unsupervised subword-phrase labeling and MTL-based text classification for the subword-phrase recognition task. Note that we assume the presence of labels for classification for a dataset. As an implementation of our framework, we employ one of the most primitive approaches, frequency-based subword-phrase labeling. A basic idea is that frequently co-occurring consecutive subwords are merged to form a subword-phrase. This idea can be implemented by various approaches, and we employ the concept of byte-pair encoding [26].

The contributions of this paper can be summarized as follows:

- **MTL-based text classification with low-cost auxiliary task preparation:** The proposed framework consists of an unsupervised labeling module and an MTL-based classification module. Existing MTL-based text classification methods assume the presence of supervision for auxiliary tasks; however, to obtain supervision for auxiliary tasks requires further human efforts and/or financial costs. In contrast, the proposed framework does not require these costs because it utilizes unsupervised subword-phrase extraction for obtaining labels to create auxiliary tasks.
- **Utilization of phrasal expression for subwords:** To the best of our knowledge, this is the first work utilizing phrasal expression for subwords (called subword-phrase). Since subwords are not necessarily semantically coherent, their phrasal expressions have not been considered for any task. In contrast, the co-occurrence of consecutive subwords (or subword-phrase) can contribute to the text classification task because such subwords may be representative to distinguish instances of a class from those of other classes. Based on this idea, this paper attempts to utilize subword-phrase in the context of MTL-based text classification.
- **Superior performance over conventional method and comparable performance with the state-of-the-art:** In the experimental evaluation on five popular text classification datasets, the proposed framework with subword-phrase recognition auxiliary task lifted up classification performance (micro and macro F-scores) from the single-task method. Compared with the state-of-the-art method (Bert-GCN [12]), the proposed framework also shows superior performance for the datasets with a more significant number of labels and comparative classification performance in the other datasets.

The rest of this paper is organized as follows. First, Section 2 introduces related work of MTL-based text classification. Second, Section 3 explains the proposed framework of MTL-based text classification with subword-phrase recognition task. Then, Section 4 showcases experimental evaluation to show the effectiveness of the proposed framework in comparison with single-task text classification baseline as well as the state-of-the-art method, and discusses the effect of subword-phrases. Finally, Section 5 concludes this paper and shows promising future directions.

2 RELATED WORK

This section introduces related literature on MTL-based text classification. According to Zhang et al. [30], MTL-based text classification methods are categorized into the following three types in terms of relationship between main task and auxiliary task.

- **Multi-Cardinality** Main and auxiliary tasks are of different datasets and in the same domain, and these tasks are also different in cardinality, meaning that text lengths, the number of classes, etc., are different. For example, Liu et al. [15] proposed MTL-based movie review classification using multiple datasets (e.g., SST dataset [27] and IMDB dataset [19]) and showed improved classification performance.
- **Multi-Domain** Main and auxiliary tasks are similar, but their domains differ. For example, Liu et al. [14] and Zhang et al. [30] examined MTL-based movie review classification with classification tasks of reviews for various products such as books and DVDs [4].
- **Multi-Objective** Main and auxiliary tasks are of different objectives. For example, Liu et al. [16] combined query classification and search result ranking in an MTL manner, and Zhang et al. [30] attempted MTL-based movie review classification (IMDB [19]) with news article classification (RN [1]) and question type classification (QC [11]) as auxiliary tasks.

In addition, MTL approaches [3] in which the main and auxiliary tasks are in the same dataset showed their effectiveness. Bi et al. [3] improved the performance of news recommendation by using MTL, which combines the news recommendation task with news article classification and named entity recognition. In the multi-label classification problem, where a single text can be classified into multiple labels, MTL-based text classification, which utilizes the relationship between labels in the same dataset, has also been proposed [31]. Zhang et al. [31] showed improved classification performance by designing an auxiliary task to learn the relationship between labels.

These studies have shown the effectiveness of combining multiple supervised learning. However, in general, creating supervised data is expensive in terms of both human efforts and financial costs; thus, lower-cost ways to design auxiliary tasks are desirable.

Self-supervised learning (SSL) is a training approach for understanding data without supervised datasets, it first hides pieces of data, and it trains a model so that the model can estimate the hidden pieces. Masked language model (MLM) is a popular SSL in the research field of natural language processing [8]. A popular pre-trained neural language model, BERT [8], is trained based on two SSL tasks: MLM and next sentence prediction. In image processing domain, DALLE-E [22] showed the significant performance of SSL, that it erases an area of an image and is trained to estimate the

erased area. The increasing attention to these models indicates the usefulness of SSL for *data understanding*.

In contrast to data understanding, text classification is a different task, that is, a supervised learning task. In other words, SSL expects models to reconstruct broken pieces of data, while supervised learning expects models to learn dedicated criteria from supervision. Therefore, task settings in SSL are not easily imported to MTL-based text classification.

The proposed framework in this paper is about creating datasets for auxiliary tasks in an unsupervised way, significantly reducing human efforts and financial costs. To the best of our knowledge, no research attempts to design auxiliary tasks of MTL-based text classification in that way. In addition, since subwords are not necessarily semantically coherent, subword-phrases have not been considered for any task. Therefore, this paper proposes a novel methodology of MTL-based text classification in two aspects: (1) low-cost auxiliary task design and (2) introduction of subword-phrases. In addition, the experimental evaluation in this paper reveals promising results for these two aspects.

3 PROPOSED FRAMEWORK

This section first explains the proposed framework of MTL-based text classification that generates subword-phrase labels for auxiliary tasks in an unsupervised manner.

3.1 Framework Overview

Figure 1 illustrates the proposed framework. It consists of two phases: unsupervised labeling and MTL-based text classification. The basic idea of the framework is that subword-phrase recognition is added as an auxiliary task for MTL-based text classification. To realize the recognition task, an unsupervised subword-phrase extraction is employed to create pseudo supervision for it. A text classifier based on the framework is trained in the following steps:

- (1) **[Input]** the text classifier receives a training set of text with classification labels and the text is tokenized into subwords using a subword-based tokenizer,
- (2) **[Phase 1]** unsupervised labeling module appends subword-phrase labels to each text in training set for the auxiliary subword-phrase recognition task, and
- (3) **[Phase 2]** the text classifier is trained in an MTL manner which is trained together with the auxiliary subword-phrase recognition task based on the appended labels.

Formally, a training set is denoted as $D = \{(T_i, y_i) \mid 1 \leq i \leq N\}$, where T_i represents a sequence of subword tokens of i -th text, y_i represents the label corresponding to i -th text, and N is the number of texts. In the first phase, the unsupervised subword-phrase labeling module receives D and performs subword-phrase extraction on subword token sequences to create another training set $D^{aug} = \{(T_i, Y_i^{aug}) \mid 1 \leq i \leq N\}$ for the auxiliary task, where Y_i^{aug} is a corresponding sequence of labels for each token in T_i . In the second phase, D and D^{aug} are passed to an MTL-based text classification module based on a pre-trained neural language model and train the text classification model in conjunction with the training subword-phrase recognition model.

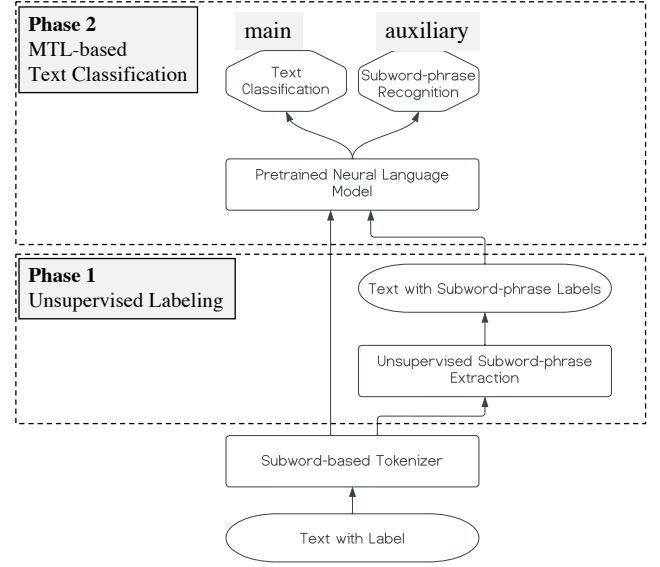


Figure 1: Proposed MTL-based Text Classification Framework. The framework accepts text with text classification labels and trains an MTL-based text classification model. The framework consists of two phases: the first phase is unsupervised labeling to the input text, and the second phase is to train the MTL-based text classification model by using the text classification labels and labels from the first phase.

3.2 Unsupervised Subword-phrase Labeling

Unsupervised subword-phrase labeling provides a label sequence corresponding with the input text sequence. This unsupervised labeling is a task formalized as follows.

- **Given:** a sequence of subword tokens T optionally with a class label y , $(T, y) \in D$
- **Generate:** a sequence of labels Y^{aug} whose length is exactly the same as the length of T

The labeling scheme is inspired by named entity recognition tasks that employs the Inside-outside-beginning (IOB2) tagging scheme [23]. IOB2 tagging is a labeling scheme that the first token of a phrase is tagged with B (beginning), the intermediate tokens of a phrase are tagged with I (inside), and tokens other than the phrase are tagged with O (outside). Besides these tags, semantic types are appended to distinguish types of phrases; for example, B-PERSON and I-PERSON respectively represent beginning and intermediate tokens of a token sequence corresponding with a person's name.

A straightforward labeling scheme for subword-phrase labeling is to treat all phrases equally. In other words, the semantic type is set to Phrase. Formally, when an n -length sequence of tokens $S = (s_1, s_2, \dots, s_n)$ has a phrase which is an m -length sub-sequence $P = (s_k, s_{k+1}, \dots, s_{k+m})$ of S where $m \leq n$, s_k is labeled as a particular type B-Phrase, the rest of tokens from s_{k+1} to s_{k+m} are labeled as I-Phrase, and other tokens $s_i \in S \setminus P$ are labeled as O.

A classification-specific labeling scheme puts different labels on subword-phrases appearing in different classes. When a subword-phrase $P = (s_k, s_{k+1}, \dots, s_{k+m})$ which is a sequence of tokens of a text and it belongs to class y , s_k is labeled as B- y and the rest of tokens from s_{k+1} to s_{k+m} are labeled as I- y . In this scheme, subword-phrases commonly appearing in different classes cannot be handled, denoting such subword-phrases as *common subword-phrases*. There are two choices to handle them, one is to eliminate them from the labeling and the other is to put a particular semantic type to label. In the latter choice, a special class label $\emptyset$ is used as a special semantic type on labeling in the IOB2 scheme, specifically, common subword-phrase P is labeled as B- $\emptyset$ for s_k and I- $\emptyset$ for other tokens.

Note that this scheme cannot distinguish the number of classes that a subword-phrase appears in texts. However, when two or more classes are similar in the domain (especially, classes are designed in fine-grained granularity), that subword-phrase appearing in texts of these classes is important to determine that a text can belong to these classes. Since this paper’s primary purpose is to recognize the usefulness of this unsupervised approach to label subword-phrases for MTL-based text classification, designing a sophisticated labeling scheme is left for future work.

3.3 MTL-based Text Classification

The proposed framework in this paper uses a pre-trained neural language model (NLM) based MTL-based text classification method. In this method, the NLM performs token encoding, and classification modules for main and auxiliary tasks are appended on top of the encoding. Therefore, NLM is the part shared among tasks and is trained in an MTL manner. The classification models are simply designed by a fully-connected layer and a softmax non-linear layer.

For the main task (i.e., text classification), a representation $\mathbf{h}^{cls}$ for a given input token sequence is obtained from NLM, and it is passed to a fully connected layer followed by a softmax layer for predicting class distribution $\hat{\mathbf{y}}^{cls}$. Formally, $\hat{\mathbf{y}}^{cls}$ for $\mathbf{h}^{cls}$ is calculated by the following equation:

$$\hat{\mathbf{y}}^{cls} = \text{softmax}(W_{cls}^\top \cdot \mathbf{h}^{cls} + \mathbf{b}^{cls}), \quad (1)$$

where W_{cls} and $\mathbf{b}^{cls}$ are a parameter matrix and bias, respectively, for the text classification task.

For the auxiliary tasks (i.e., subword-phrase recognition), a representation $\mathbf{h}_j^{spr}$ for j -th token of a given input sequence is obtained from NLM. It passes to a fully connected layer followed by a softmax layer for predicting token label distribution $\hat{\mathbf{y}}^{spr}$. Formally, $\hat{\mathbf{y}}_j^{spr}$ for $\mathbf{h}_j^{spr}$ is calculated by the following equation:

$$\hat{\mathbf{y}}_j^{spr} = \text{softmax}(W_{spr}^\top \cdot \mathbf{h}_j^{spr} + \mathbf{b}^{spr}), \quad (2)$$

where W_{spr} and $\mathbf{b}^{spr}$ are a parameter matrix and bias, respectively, for the subword-phrase recognition task.

These main and auxiliary tasks are multi-class classification tasks; therefore, it is straightforward to use the cross entropy loss as a loss function. The loss L_{cls} for the text classification task is calculated by the following equation:

$$L_{cls} = - \sum_{i=1}^N \sum_{c \in C} y_{i,c} \log \hat{y}_{i,c}^{cls}, \quad (3)$$

where N is the number of training sample texts, C is a set of classes, $y_{i,c} \in \{0, 1\}$ is a true label for i -th text where $y_{i,c} = 1$ if the true label of the text is c , 0 otherwise, and $\hat{y}_{i,c}^{cls}$ is predicted probability of class c for the text.

Similarly, the loss L_{spr} for the subword-phrase recognition task is calculated by the following equation:

$$L_{spr} = - \sum_{i=1}^N \sum_{j=1}^{M_i} \sum_{c \in C} y_{i,j,c} \log \hat{y}_{i,j,c}^{cls}, \quad (4)$$

where N is the number of training sample texts, M_i is the number of tokens in i -th text, C is a set of classes, $y_{i,j,c} \in \{0, 1\}$ is a true label for j -th token of i -th text where $y_{i,j,c} = 1$ if the true label of the token is c , 0 otherwise, and $\hat{y}_{i,j,c}^{spr}$ is predicted probability of class c for the token.

To train both tasks simultaneously, feedback from results on these tasks is fed to the NLM model to fine-tune its parameters. Therefore, joint loss L_{joint} of losses for these tasks are calculated by the following equation and used for parameter optimization.

$$L_{joint} = L_{cls} + L_{spr} \quad (5)$$

Note that the weighting scheme in MTL approaches to involve the importance of individual tasks has been studied [20, 25]. Although taking the weighting scheme into the proposed framework is promising, the purpose of this paper is to show the capability of the MTL-based text classification in conjunction with subword-phrase recognition which labels for auxiliary tasks are created in an unsupervised manner. Therefore, taking the weighting scheme to the proposed framework is a promising future step.

4 EXPERIMENTAL EVALUATION

To evaluate the proposed framework, we conducted an experimental evaluation to answer the following questions:

- **Q1:** Does the MTL-based text classification method in which an auxiliary task is created in an unsupervised manner improve the classification performance compared with the single-task text classification method?
- **Q2:** Can the MTL-based text classification outperform the state-of-the-art (SOTA) text classification method?
- **Q3:** Does subword-phrase contribute to text classification?

The rest of this section is organized as follows: Section 4.1 introduces the implementation of the proposed framework, Section 4.2 explains the SOTA text classification method for comparison, Section 4.3 describes experimental settings, Section 4.4 showcases experimental results, Section 4.5 analyses subword-phrases extracted and used for the subword-phrase recognition auxiliary task, and Section 4.6 introduces remarks of the experiments by answering these questions.

4.1 Implementation of the Proposed Framework

In this experiment, we implemented a simple frequency-based subword-phrase extraction method, and the labeling scheme for the extracted subword-phrase was the classification-specific labeling scheme. The frequency-based method expects that frequently co-occurring subwords compose regular textual expressions for each class. In order to control the number of subword-phrases, we

utilized Byte Pair Encoding algorithm (BPE) [26]. BPE concatenates consecutive tokens if they are frequently co-occurring in a corpus, and repeats this concatenation until the number of unique tokens becomes the expected number. The ability to control the number of subword-phrases was suitable for this experiment because subword-phrase were newly proposed in this paper; therefore, we needed to try several variations, which were realized by creating different numbers of subword-phrases.

In general, the numbers of texts are skewed among classes; that is, the number of texts of a class is quite large, while that of other classes is very few. This also affected the extraction of subword-phrases; therefore, in this experiment, the extraction mentioned above was applied for each set of texts of class. Specifically, we extracted n subword-phrases for each class. n was chosen from $\{10, 100, 1000, 10000\}$ to achieve the best classification performance on the validation data.

4.2 Comparison Method: BertGCN

BertGCN [12] is a SOTA method of text classification that combines a pre-trained NLM with inductive learning of graph neural networks (GNNs). BertGCN follows TextGCN [29] by constructing a graph of co-occurrence relations between texts and words and between words and words. In BertGCN, vectors of vertices are initialized using the pre-trained NLM. These vectors are updated through graph convolutional neural network (GCN) in order to involve the relationships between texts and words. Based on the updated vectors, BertGCN performs text classification by adding a fully connected layer followed by a softmax layer. In addition, [12] reported that integrating the output of the NLM-based classification model and that of BertGCN can improve classification performance. Specifically, the linear sum of the predicted class distributions Z_{GCN} and Z_{NLM} obtained from BertGCN and the classifier using NLM, respectively, as seen in the following equation:

$$Z = \lambda \cdot Z_{GCN} + (1 - \lambda) \cdot Z_{NLM}, \quad (6)$$

where $\lambda \in [0, 1]$ is the weight for BertGCN classification. In this experiment, $\lambda = 0.7$ was used since [12] reported it was the best.

4.3 Settings

Datasets. For the evaluation, the following popular datasets in the text classification task are used.

- **MR (Movie Review):** Dataset of movie reviews categorized into binary sentiment classes (i.e., positive and negative)
- **20NG (20 Newsgroups):** Dataset of news texts categorized into 20 categories
- **R8:** Dataset of news articles from Reuters-21578¹ limited to selected 8 classes
- **R52:** Dataset of news articles from Reuters-21578 limited to selected 52 categories
- **OHS (Ohsumed):** Dataset of medical abstracts which are categorized into 23 medical concepts called MESH categories

The statistics of the dataset are shown in Table 1. As the table shows, datasets with different classes and variations in the number of instances per class (the standard deviation (Std.) of the number of

Table 1: Statistics of datasets. The number of instances in train-valid-test splits, the number of classes, and average (Avg.) and standard deviation (Std.) of the numbers of instances across classes.

	MR	20NG	R8	R52	OHS
#Train	6,398	10,183	4,937	5,879	3,022
#Valid	710	1,131	548	653	335
#Test	3,554	7,532	2,189	2,568	4,043
#Class	2	20	8	52	23
Avg. #Instances/Class	5,331	942	959	175	321
Std. #Instances/Class	0	94	1,309	613	305

instances within a class) were used in the experiment. These various datasets were expected to reveal the advantages and disadvantages of the proposed method.

Metrics. The evaluation metric is F -score which is the harmonic mean of precision and recall scores as shown below.

$$Pre = \frac{TP}{TP + FP}, \quad Rec = \frac{TP}{TP + FN}, \quad F = \frac{2 \cdot Pre \cdot Rec}{Prec + Rec}$$

The precision (denoted Pre) is a ratio of the number of true positives (TP) over the number of instances estimated as positive (i.e., $TP + FP$; where FP is the number of false positives). The recall (denoted Rec) is a ratio of TP over the number of positive instances in the evaluation set (i.e., $TP + FN$; where FN is the number of false negatives). To observe various aspects for evaluation, micro and macro averages of F -scores are used in this experiment. The micro average of F -scores, F_{micro} , is the instance-level average of F -score, and the macro average, F_{macro} , is the class-level average of F -scores.

Parameters. For the base model in the proposed method and BertGCN, we used the RoBERTa-base model [17] available at Huggingface². BertGCN with RoBERTa model was called RoBERTaGCN in this experiment. In this paper, the effect of common subword-phrases was also evaluated, therefore, the proposed method had two variation: one included common subword-phrases (denoted Proposed w/ cmn) and the other excluded them (denoted Proposed w/o cmn). In addition, as a baseline method, we also used a single-task text classification method based on RoBERTa. The baseline method was implemented by adding a fully connected layer and a softmax layer on top of RoBERTa, which is equivalent to Eq. 1 with the loss function shown in Eq. 3. The only difference between the proposed method and the baseline method was the number of tasks on top of RoBERTa, therefore, the comparison between them was expected to reveal the effectiveness of MTL-based text classification. These models were optimized using AdamW optimizer (Adam optimizer [9] with decoupled weight decay regularization) [18]. Basically, experiments were conducted with 100 epochs, 64 batch size, and a maximum token length of 256. Only the experiment for RoBERTaGCN was conducted with 128 batch size and a maximum token length of 128, which gave better results than the aforementioned hyper-parameters.

¹Reuters-21578, <http://www.daviddlewis.com/resources/testcollections/reuters21578/>, viewed on August 4, 2022

²<https://huggingface.co/roberta-base>

Table 2: Evaluation Results: The best score in each column (i.e., dataset) is bold-faced. RoBERTaGCN is the SOTA text classification method and Baseline is the single-task text classification based on RoBERTa model. The proposed method has two variations: one, denoted Proposed w/ cmn, includes common subword-phrases in the labeling scheme, and the other, denoted Proposed w/o cmn, excludes them. (a) and (b) showcase results of F_{micro} and F_{macro} , respectively.

(a) F_{micro}						(b) F_{macro}					
Model	MR	20NG	R8	R52	OHS	Model	MR	20NG	R8	R52	OHS
RoBERTaGCN	0.880	0.894	0.979	0.944	0.736	RoBERTaGCN	0.880	0.861	0.925	0.756	0.605
Baseline (RoBERTa)	0.881	0.831	0.977	0.962	0.690	Baseline (RoBERTa)	0.881	0.825	0.943	0.836	0.594
Proposed w/ cmn	0.860	0.850	0.978	0.967	0.704	Proposed w/ cmn	0.860	0.845	0.947	0.841	0.610
Proposed w/o cmn	0.866	0.851	0.979	0.969	0.711	Proposed w/o cmn	0.866	0.845	0.955	0.851	0.637

4.4 Results

Table 2 shows the experimental results of F_{micro} (Table 2(a)) and F_{macro} (Table 2(b)), and showcases the following three observations:

- (1) the proposed method performed better than the baseline method in both metrics except the simple binary classification on the MR dataset,
- (2) the common subword-phrases slightly decreased the classification performance of the proposed method, and
- (3) the proposed method outperformed RoBERTaGCN performed three out of five datasets and performed comparably with the SOTA in terms of the F_{macro} metric (Table 2(b)),

The comparison between the proposed method and the baseline method in both F_{micro} and F_{macro} revealed the effectiveness of the MTL-based approach in which auxiliary task is systematically constructed. In addition to insights from the existing literature that MTL-based approaches using auxiliary tasks with supervision are effective, this experiment showcased the effectiveness of an MTL approach in which training data for an auxiliary task is generated in an unsupervised manner. Based on the result, low-cost auxiliary tasks for MTL-based text classification have become promising.

The results of MR and R8 datasets showed comparable performances between the proposed and the baseline methods. These datasets were composed of smaller numbers of classes. These results suggest that the proposed method did not perform effectively when the number of classes is small.

The comparison between two variations of the proposed method (one includes common subword-phrases and the other exclude them) showed that the exclusion of the common subword-phrases slightly increased the classification performance. The common subword-phrases were not class-specific phrases, therefore, excluding them enabled the proposed method improving the proposed method. Note that the proposed method with common subword-phrases still outperformed the baseline method. Therefore, MTL-based text classification with subword-phrase recognition auxiliary task itself was valuable to improve the classification performance. In consequence, choices of subword-phrases for the auxiliary task affected the classification performance.

A notable fact from the results in R52 and OHS datasets was the proposed method achieved significantly better than RoBERTaGCN in F_{macro} . At the same time, the proposed method was also more accurate than RoBERTaGCN in terms of F_{micro} . Recalling the statistics of these datasets from Table 1, the numbers of its classes are larger than those of other datasets, and the number of instances

per class is highly skewed. These facts indicate that the proposed method is good at highly skewed datasets. Through 20NG dataset had similar number of classes to OHS dataset and smaller skewness than OHS dataset, the performance in F_{micro} and F_{macro} of the proposed method was worse than RoBERTaGCN. In consequence, the proposed method performed better than the SOTA method when datasets were composed of large number of classes and highly skewed in the number of instances across classes.

4.5 Analysis of Subword-phrases

Recalling that the unsupervised subword-phrase extraction module was implemented by a frequency-based approach and it extracted n subword-phrases for each class, the unique subword-phrases in each class were labeled with class label (i.e., B- y and I- y), and the common subword-phrases across classes were labeled with special semantic type (i.e., B- $\emptyset$ and I- $\emptyset$).

To realize the effect of the subword-phrase recognition task as an auxiliary task, Table 3 shows the top 10 frequent subword-phrases for each class in the R52 dataset. Note that the special symbol “Ġ” in the subword-phrase is assigned by the RoBERTa-base tokenizer, and it represents the starting position of the word when it is split by a space delimiter. Table 3 shows that the selected classes bop (BOP; balance of payments), money-fx, and orange are among 52 classes and common subword-phrases. These classes are categories of financial news articles from different perspectives, therefore, they have many subword-phrases related to finance. For example, “money supply”, “borrowings” and “business loan” appeared in the bop class, petroleum-related subword-phrases like “opec” (OPEC; Organization of the Petroleum Exporting Countries) and “mln bpd” (million barrels per day) appeared in the money-fx class, and “consumer price index” appeared in the orange class. These subword-phrases captured the characteristics related to these financial categories.

The common subword-phrases include more general phrases. For instance, “reuter”, “january”, “spokesman said”, and “pct” (percent) appear in news articles in wide range of classes. They are indeed not relevant to specific classes. In this paper, there were two choices of handling the common subword-phrases: one was to put a particular semantic type and the other was to exclude them, and the experimental results showcased the latter is better. While, the proposed method with common subword-phrases was still better than the baseline method. Therefore, the discussion on treating common subword-phrases needs more investigation, suggesting that the subword-phrase extraction module is required to extract

Table 3: Top-10 frequent subword-phrases of selected classes and common across classes in the R52 dataset. Subword-phrases are represented space-concatenation of subwords. Ġ indicates the starting position of word obtained from RoBERTa tokenizer.

Class Name				
	bop	money-fx	orange	Common Subword-phrase
1	Ġmoney Ġsupply	Ġo pec	Ġconsumer Ġprices	Ġre uter
2	ĠmIn Ġdlrs Ġbillion Ġdlrs	Ġb pd	Ġconsumer Ġprice Ġindex	Ġfe b
3	Ġweek Ġended	Ġec uador	Ġstatistics Ġinstitute	Ġl me
4	Ġborrow ings	Ġcrude Ġoil	Ġrose Ġpct Ġfebruary	Ġjan uary
5	Ġbusiness Ġloans	Ġsa udi	Ġpct Ġmarch	Ġfeb ruary
6	Ġmoney Ġsupply Ġrises	ĠmIn Ġbpd	Ġcost Ġiving	Ġspokesman Ġsaid
7	Ġfed Ġsays	ĠmIn Ġbarrels	Ġpct Ġcompared	Ġp ct
8	Ġbillion Ġdlrs Ġbillion Ġdlrs	Ġsaudi Ġarab	-	Ġm ln
9	Ġweek Ġfed Ġsays	Ġdlrs Ġbarrel	-	Ġseason ally
10	Ġpct Ġbillion	ĠsaudiĠarab ia	-	Ġseasonally Ġadjusted

class-specific subword-phrases. Seeking approaches realizing this has remained for the future direction.

4.6 Remark

This section summarizes findings from the experiment by answering the questions and introduces the limitations of the proposed method revealed by the experiment.

Q1: Does the MTL-based text classification method in which an auxiliary task is created in an unsupervised manner improve the classification performance compared with the single-task text classification method? — Yes, it does. The proposed method clearly outperformed the baseline method when the number of classes of a dataset was large, and they are comparable when that number is small. On the other perspective, datasets with that small number of classes were also less skewed in the number of instances per class. Therefore, the frequency-based subword-phrase extraction for constructing auxiliary task was suitable for datasets with large number of classes and the number of instances per class was skewed. A promising outcome is that a recognition auxiliary task which (pseudo) supervision is generated in an unsupervised manner is effective in MTL-based classification. Therefore, this outcome opens up new possibility to construct auxiliary tasks for MTL-based classification methods on classification tasks other than text classification.

Q2: Can the MTL-based text classification outperform the state-of-the-art (SOTA) text classification method? — Yes, but in limited cases. The proposed method was superior to the SOTAMethod, RoBERTaGCN, for R52 and OHS datasets, which contained large number of classes and the number of instances per class was skewed. A promising direction to overcome the inferiority of the proposed method in the other datasets is to utilized RoBERTaGCN as a base model of the proposed method.

Q3: Does subword-phrase contribute to text classification? Yes, it does. The subword-phrase recognition task as an auxiliary task improves text classifications in various datasets. The extracted subword-phrases were class-specific as expected, while the usage of common subword-phrases is remained challenging. This

paper employs a simple labeling approach that subword-phrases appearing in texts of two or more classes were treated as common subword-phrases. This simple approach did not flexibly handle the commonality of subword-phrases. Therefore, it is necessary to seek a suitable labeling scheme that can handle different commonalities. A promising outcome is the usage of phrasal expressions for subwords, which has not been paid much attention by any research.

Limitations of the Proposed Method. These experiments revealed the limitations of the proposed methods. The following introduces these limitations and promising future directions to overcome them.

- The proposed method did not perform well on the datasets having a balanced number of instances per class, in other words, less skewed. This can be caused by implementing the unsupervised subword-phrase extraction module, which was a simple frequency-based approach. Dataset with a smaller number of instances was *forced* to extract subword-phrases even though many subwords were not frequently occurring. Since this paper showcases the effectiveness of the proposed method, the next step to improve the proposed method is to sophisticate that module by incorporating the valuable insights obtained from natural language processing researches.
- As shown in Table 3, subword-phrases typically composed a word or phrase in general sense. This can be caused by the frequency-based method for subword-phrase extraction, that is, when a word is frequently occurring in a corpus, subwords which compose that word are also frequently occurring. As a result, these subwords are selected to compose a subword-phrase. Therefore, this is a limitation for the frequency-based subword-phrase extraction method. From the other perspective, this fact can be a suggestion for developing other subword-phrase extraction methods which are based on word-level phrasal expressions. For example, finding important words such that those words represents documents has conducted in the research field of information retrieval. Such words can be subword-phrases in the

proposed method perspective. Therefore, these facts suggest a future direction to utilize word or word-level phrase research insights.

5 CONCLUSION

This paper proposes a multi-task learning-based text classification framework that creates supervision for auxiliary tasks in an unsupervised way to reduce human efforts and financial costs for training classification models. This paper also seeks the possibility of phrasal expressions of subwords (called subword-phrase) to utilize subword-based neural language pre-trained models (like BERT). As an implementation of the proposed framework, we employed a frequency-based extraction method for subword-phrases. The experimental evaluation of the five popular datasets for text classification showcases the effectiveness of the involvement of the subword-phrase recognition as an auxiliary task. It also shows comparative results with the state-of-the-art method.

Promising outcomes of this paper are two folds: (1) a recognition auxiliary task which pseudo supervision is generated in an unsupervised manner is effective in MTL-based classification, and this opens up new possibility to construct auxiliary tasks for MTL-based classification methods on classification tasks other than text classification; and (2) phrasal expressions for subwords (subword-phrase) can be useful in text classification.

For the future direction, we first seek other unsupervised subword-phrase extraction methods. The usage of subword-phrase is not necessarily close to ordinary word-based phrases, and there is no previous insight into subword-phrases. Therefore, we are going to seek this. Second, the labeling scheme for subword-phrase recognition is extended to well-handle the common subword-phrases among classes. Last, weighting schemes for multi-task learning [20, 25] are going to be enhanced in the proposed framework to further improve classification performance.

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REFERENCES

- [1] Chidanand Apté, Fred Damerau, and Sholom M. Weiss. 1994. Automated Learning of Decision Rules for Text Categorization. *ACM Trans. Inf. Syst.* 12, 3 (1994), 233–251.
- [2] Alberto Benayas, Reyhane Hashempour, Damian Rumble, Shoaib Jameel, and Renato Cordeiro De Amorim. 2021. Unified Transformer Multi-Task Learning for Intent Classification With Entity Recognition. *IEEE Access* 9 (2021), 147306–147314.
- [3] Qiwei Bi, Jian Li, Lifeng Shang, Xin Jiang, Qun Liu, and Hanfang Yang. 2022. MTRec: Multi-Task Learning over BERT for News Recommendation. In *ACL 2022*. 2663–2669.
- [4] John Blitzer, Mark Dredze, and Fernando Pereira. 2007. Biographies, Bollywood, Boom-boxes and Blenders: Domain Adaptation for Sentiment Classification. In *ACL 2007*. 440–447.
- [5] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. In *NeurIPS 2020*.
- [6] Rich Caruana. 1997. Multitask Learning. *Machine Learning* 28, 1 (1997), 41–75.
- [7] Ona de Gibert, Naiara Pérez, Aitor García Pablos, and Montse Cuadros. 2018. Hate Speech Dataset from a White Supremacy Forum. In *ALW@EMNLP 2018*. 11–20.
- [8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT 2019*. 4171–4186.
- [9] Diederik P. Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *ICLR 2015*.
- [10] Qian Li, Hao Peng, Jianxin Li, Congying Xia, Renyu Yang, Lichao Sun, Philip S. Yu, and Lifang He. 2022. A Survey on Text Classification: From Traditional to Deep Learning. *ACM Trans. Intell. Syst. Technol.* 13, 2 (2022), 31:1–31:41.
- [11] Xin Li and Dan Roth. 2002. Learning Question Classifiers. In *COLING 2002*.
- [12] Yuxiao Lin, Yuxian Meng, Xiaofei Sun, Qinghong Han, Kun Kuang, Jiwei Li, and Fei Wu. 2021. BertGCN: Transductive Text Classification by Combining GNN and BERT. In *ACL-IJCNLP 2021*. Online, 1456–1462.
- [13] Marco Lippi, Przemysław Palka, Giuseppe Contissa, Francesca Lagioia, Hans-Wolfgang Micklitz, Giovanni Sartor, and Paolo Torroni. 2019. CLAUDETTE: an automated detector of potentially unfair clauses in online terms of service. *Artif. Intell. Law* 27, 2 (2019), 117–139.
- [14] Pengfei Liu, Xipeng Qiu, and Xuanjing Huang. 2016. Deep Multi-Task Learning with Shared Memory for Text Classification. In *MNLP 2016*. 118–127.
- [15] Pengfei Liu, Xipeng Qiu, and Xuanjing Huang. 2016. Recurrent Neural Network for Text Classification with Multi-Task Learning. In *IJCAI 2016*. 2873–2879.
- [16] Xiaodong Liu, Jianfeng Gao, Xiaodong He, Li Deng, Kevin Duh, and Ye-Yi Wang. 2015. Representation Learning Using Multi-Task Deep Neural Networks for Semantic Classification and Information Retrieval. In *NAACL-HLT 2015*. 912–921.
- [17] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *CoRR abs/1907.11692* (2019).
- [18] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6–9, 2019*. OpenReview.net. <https://openreview.net/forum?id=Bkg6RiCqY7>
- [19] Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. 2011. Learning Word Vectors for Sentiment Analysis. In *ACL 2011*. 142–150.
- [20] Yuren Mao, Zekai Wang, Weiwei Liu, Xuemin Lin, and Pengtao Xie. 2022. MetaWeighting: Learning to Weight Tasks in Multi-Task Learning. In *ACL 2022*. 3436–3448.
- [21] Shervin Minaee, Nal Kalchbrenner, Erik Cambria, Narjes Nikzad, Meysam Chenaghlu, and Jianfeng Gao. 2021. Deep Learning-based Text Classification: A Comprehensive Review. *ACM Comput. Surv.* 54, 3 (2021), 62:1–62:40.
- [22] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. 2021. Zero-Shot Text-to-Image Generation. In *ICML 2021*. 8821–8831.
- [23] Lance Ramshaw and Mitch Marcus. 1995. Text Chunking using Transformation-Based Learning. In *Third Workshop on Very Large Corpora*.
- [24] Fabrizio Sebastiani. 2002. Machine Learning in Automated Text Categorization. *ACM Comput. Surv.* 34, 1 (mar 2002), 1–47. <https://doi.org/10.1145/505282.505283>
- [25] Ozan Sener and Vladen Koltun. 2018. Multi-Task Learning as Multi-Objective Optimization. In *NeurIPS 2018*. 525–536.
- [26] Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Neural Machine Translation of Rare Words with Subword Units. In *ACL 2016*.
- [27] Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. 2013. Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank. In *EMNLP 2013*. 1631–1642.
- [28] Chaochen Wu, Guan Luo, Chao Guo, Yin Ren, Anni Zheng, and Cheng Yang. 2020. An attention-based multi-task model for named entity recognition and intent analysis of Chinese online medical questions. *Journal of Biomedical Informatics* 108 (2020), 103511.
- [29] Liang Yao, Chengsheng Mao, and Yuan Luo. 2019. Graph Convolutional Networks for Text Classification. In *AAAI 2019*. 7370–7377.
- [30] Honglun Zhang, Liqiang Xiao, Yongkun Wang, and Yaohui Jin. 2017. A Generalized Recurrent Neural Architecture for Text Classification with Multi-Task Learning. In *IJCAI 2017*. 3385–3391.
- [31] Ximing Zhang, Qian-Wen Zhang, Zhao Yan, Ruifang Liu, and Yunbo Cao. 2021. Enhancing Label Correlation Feedback in Multi-Label Text Classification via Multi-Task Learning. In *ACL/IJCNLP 2021*. 1190–1200.
- [32] Yongzheng Zhang, Nur Zincir-Heywood, and Evangelos Milios. 2005. Narrative Text Classification for Automatic Key Phrase Extraction in Web Document Corpora. In *Proceedings of the 7th Annual ACM International Workshop on Web Information and Data Management (Bremen, Germany) (WIDM '05)*. 51–58. <https://doi.org/10.1145/1097047.1097059>